Causal Inference with Ordinal Outcomes: Flexible Estimation Framework Using Normalizing Flows

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Abstract

Ordinal outcomes, such as Likert-type survey responses, are ubiquitous in political science, yet they pose distinctive challenges for causal inference. These outcomes are routinely analyzed by assigning numerical scores and applying linear models, despite longstanding concerns that ordinal responses do not identify treatment effects on an underlying cardinal scale without additional assumptions. Building on the classical identification result for ordinal outcomes and existing recommendations to report probability-based quantities, I develop a flexible likelihood-based framework for estimating distributional causal effects, including category-specific and cumulative treatment effects, which are invariant to admissible transformations of the latent scale. Existing approaches typically estimate these quantities using parametric ordered logit or ordered probit models, which rely on restrictive assumptions about the latent error distribution and index specification. This paper proposes a flexible likelihood-based estimation framework for ordinal causal inference using conditional normalizing flows. The proposed estimator flexibly learns the conditional distribution underlying observed ordinal responses without imposing a parametric latent error distribution, while preserving the likelihood-based structure familiar from classical ordered response models. The same framework accommodates both structured latent-index models and fully nonparametric conditional distribution estimators, allowing researchers to balance interpretability and modeling flexibility within a unified estimation procedure. Monte Carlo simulations show that the proposed estimator maintains low bias across a broad range of latent error distributions and nonlinear response mechanisms under which conventional ordered response models perform poorly. Replications of two published survey experiments demonstrate that flexible distributional estimation can materially alter substantive conclusions about experimental treatment effects.

1. Introduction

Causal inference has become a central organizing principle in political science. Experimental and quasi-experimental designs, including survey experiments, difference-in-differences, and regression discontinuity designs, are now standard empirical strategies for studying the effects of policies, institutions, and information. Survey experiments are particularly prominent because random assignment provides a transparent basis for causal identification. Between 2021 and 2024, roughly 20% of articles published in the *American Political Science Review*, *American Journal of Political Science*, and *Journal of Politics* employed a survey experiment as part of their empirical strategy.

Many of these studies investigate outcomes that are inherently latent, including support for redistribution (Alt and Iversen, 2017; Magni, 2021), evaluations of political leaders (Canes-Wrone and De Marchi, 2002; Kriner and Schwartz, 2009), and foreign policy preferences (Tomz and Weeks, 2020). Although these concepts are commonly viewed as lying on an underlying continuous attitude scale, they are almost always observed through ordinal response categories such as Likert-type scales ranging from *strongly disagree* to *strongly agree*. Consequently, causal inference in political science routinely relies on outcomes that preserve the ordering of responses but not the distances between them.

Applied researchers typically analyze ordinal outcomes in one of three ways. The most common approach assigns numerical scores to response categories and estimates treatment effects using difference-in-means or ordinary least squares, implicitly treating ordinal responses as cardinal. Another common strategy collapses multiple categories into a binary outcome before estimation, sacrificing information about intermediate responses. A third approach fits ordered response models, such as ordered logit or ordered probit, which explicitly account for the ordinal nature of the data through a latent-variable representation. Each approach reflects a different set of assumptions about the relationship between the observed ordinal responses and the underlying latent attitude.

This paper begins from a classical observation in the ordinal response literature: latent treatment effects are identified only up to location and scale transformations of the underlying latent variable. Consequently, treatment effects defined through arbitrary numerical codings of ordinal responses generally require additional measurement assumptions linking the observed categories to a cardinal latent scale (Schröder and Yitzhaki, 2017; Bond and Lang, 2018; Bloem, 2022). This observation motivates a focus on distributional causal quantities, such as category-specific and cumulative treatment effects, which are defined directly on the observed ordinal responses and do not rely on assumptions about the cardinal spacing between adjacent categories.

This perspective complements recent recommendations in political methodology to summarize ordinal treatment effects using predicted probabilities rather than latent coefficients (Hanmer and Ozan Kalkan, 2013). Rather than revisiting how ordered response models should be interpreted, this paper asks how these probability-based causal quantities should be estimated when the assumptions underlying conventional ordered response models are difficult to justify. The methodological focus therefore shifts from interpreting latent coefficients to flexibly estimating the conditional ordinal response distribution. Existing semiparametric estimators relax some of these assumptions but remain difficult to apply with rich covariate structures or nonlinear response mechanisms that are common in modern survey research.

Estimating these distributional causal effects requires modeling the conditional distribution of ordinal responses. Ordered logit and ordered probit remain the workhorses of applied political science because they efficiently exploit the ordering of response categories, admit interpretable latent-variable representations, and often perform well when their assumptions are approximately satisfied. Their validity, however, depends on assumptions about the latent error distribution and index specification that may be difficult to justify in many survey experiments. Political attitudes are frequently polarized or clustered, producing multimodal latent distributions, while persuasive treatments often affect moderate respondents differently from strong supporters, generating nonlinear treatment-response relationships. In such settings, misspecification can distort estimated category probabilities and cumulative treatment effects, even when the experimental design itself identifies the causal effect. The practical challenge is therefore not causal identification, but flexible estimation of the conditional ordinal response distribution.

This paper develops a flexible likelihood-based framework for estimating causal effects with ordinal outcomes. The proposed estimator employs conditional normalizing flows to model the conditional response distribution without imposing a fixed parametric latent error distribution. Unlike existing semiparametric estimators, the framework remains computationally feasible with high-dimensional covariates while preserving exact likelihood-based inference. A single estimation procedure spans classical latent-index models, generalized latent-index specifications, and fully nonparametric conditional response distributions, allowing researchers to balance structural assumptions and modeling flexibility while targeting the same probability-based causal quantities.

I evaluate the proposed framework through Monte Carlo simulations designed to reflect empirically realistic survey settings, including polarized latent attitudes, nonlinear treatment-response relationships, and non-Gaussian latent error distributions. Across these settings, the proposed estimator maintains low bias while conventional ordered response models exhibit substantial distortions in estimated probability effects. I then revisit the survey experiments of Tomz and Weeks (2020) and Mattingly et al. (2025), showing that flexible estimation of the ordinal response distribution can materially alter substantive conclusions regarding the magnitude and distribution of treatment effects.

The remainder of the paper proceeds as follows. Section 2 formalizes causal inference with ordinal outcomes and motivates distributional causal estimands. Section 3 reviews existing estimation approaches and introduces the proposed likelihood-based framework. Section 4 evaluates finite-sample performance through Monte Carlo simulations. Section 5 presents the empirical applications. Section 6 concludes.

2. Causal Inference with Ordinal Outcomes

This section develops the conceptual framework underlying the remainder of the paper. I begin by describing ordinal survey responses as measurements of an underlying latent attitude using the classical threshold-crossing model. I then revisit a well-known identification result showing that latent treatment effects are identified only up to location and scale, and discuss its implications for causal inference with ordinal outcomes. Finally, I introduce distributional causal estimands that are directly identified from observed ordinal responses and motivate the estimation framework developed in the next section.

2.1. Latent Outcomes, Ordinal Responses, and Causal Effects

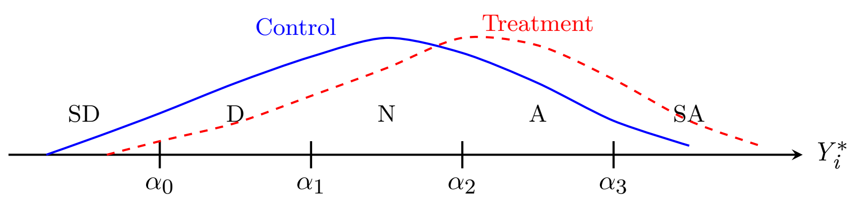


FIGURE 1. Ordinal responses arise by partitioning an underlying latent attitude into ordered response categories. The observed data identify how treatment changes probabilities of falling into response categories, but not the absolute scale of the latent attitude.

Many quantities of interest in political science, such as support for redistribution (Alt and Iversen, 2017; Magni, 2021), immigration preferences (Mayda and Rodrik, 2005), trust in institutions, and foreign policy attitudes (Scheve and Slaughter, 2001; Tomz and Weeks, 2020), are naturally viewed as continuous latent constructs. Survey respondents, however, rarely report these attitudes on a continuous scale. Instead, researchers observe ordinal responses, such as Likert items ranging from *strongly disagree* to *strongly agree*, ordered approval categories, or frequency scales. Consequently, empirical analyses are conducted using outcomes that preserve the ordering of respondents' attitudes but not the distances between adjacent response categories.

Following the classical ordered response framework, let Y_i^* denote respondent i's latent attitude and let $Y_i \in 0, 1, \ldots, J$ denote the observed ordinal response. The observed response is generated through a threshold-crossing mechanism,

$$Y_i = j \quad \iff \quad \alpha_{j-1} < Y_i^* \leq \alpha_j, \qquad j = 0, \ldots, J,$$

where

$$-\infty = \alpha_{-1} < \alpha_0 < \cdots < \alpha_J = \infty$$

are unknown thresholds partitioning the latent scale. Figure 1 illustrates this measurement process. Throughout the paper I assume respondents share a common reporting function. Throughout the paper I assume respondents share a common reporting function. Section B of the Online Appendix discusses extensions allowing reporting thresholds to vary across respondents following generalized ordered response models (Williams, 2006). Alternative survey designs, such as anchored vignettes (King et al., 2004; King and Wand, 2007), provide complementary approaches for addressing heterogeneous reporting behavior.

The threshold-crossing representation should be interpreted as a measurement model rather than a structural model. It formalizes the idea that ordinal responses preserve the ordering of the underlying attitudes while discarding information about the distances between adjacent categories. Consequently, the observed responses provide only partial information about the latent construct: they reveal which interval of the latent scale a respondent occupies, but not the respondent's exact location within that interval.

To describe causal effects, let $Y_i^*(d)$ denote the latent potential outcome under treatment level d, and let

$$Y_i(d) = g\left(Y_i^*(d)\right)$$

denote the corresponding observed ordinal potential outcome. Throughout the paper I adopt the standard assumptions of SUTVA, conditional ignorability, and positivity (Rubin, 1974). If the latent outcome Y_i^* were directly observed, these assumptions would suffice to identify conventional causal estimands such as

$$ATE^* = \mathbb{E} = \left[Y_i^*(1) - Y_i^*(0)\right].$$

In practice, however, researchers observe only the ordinal response Y_i. The central question is therefore not whether causal effects are identified, but which causal quantities remain identified once the underlying latent attitude is observed only through ordered response categories. The next subsection revisits the classical identification result for ordinal response models and uses it to distinguish between causal quantities that require additional measurement assumptions and those that do not.

2.2. Identification of Causal Effects from Ordinal Outcomes

The threshold-crossing framework provides a useful representation of ordinal outcomes, but it also reveals a fundamental limitation of what can be learned from ordinal data. A classical result in the ordered response literature is that latent-variable models are identified only up to location and scale. Although this normalization is standard in econometrics and psychometrics, its implications for causal inference have received comparatively little attention in applied political science. In particular, survey experiments frequently interpret treatment effects estimated from numerically coded ordinal responses as effects on an underlying latent attitude without making explicit the measurement assumptions required for such an interpretation.

To formalize this point, suppose the latent outcome satisfies

$$Y_i^* = h_\theta(X_i) + D_i^\top \tau + \varepsilon_i, \tag{1}$$

where $h(X_i, \beta)$ denotes the systematic component of the latent outcome, τ is the treatment effect on the latent scale, and ε_i has cumulative distribution function F. Combining this latent-index specification with the threshold-crossing measurement model implies

$$\begin{aligned} P(Y_i = j \mid X_i, D_i) &= F\left(\alpha_j - h(X_i, \beta) - D_i^\top \tau\right) \\ &= F\left(\alpha_{j-1} - h(X_i, \beta) - D_i^\top \tau\right). \end{aligned}$$

Now consider any constants $a \in \mathbb{R}$ and $c > 0$, and define a transformed latent outcome

$$\tilde{Y}_i^* = a + c Y_i^*,$$

together with transformed thresholds

$$\tilde{\alpha}_j = a + c\alpha_j.$$

Because both the latent outcome and the thresholds are transformed simultaneously, the probabilities of the observed ordinal responses remain unchanged.

Theorem 1 (Location and Scale Nonidentification). *Under the threshold-crossing latent-variable model, any positive affine transformation of the latent outcome and thresholds generates the same distribution of the observed ordinal responses. Consequently, treatment effects on the latent outcome are identified only up to location and scale.*

Theorem 1 should not be interpreted as a failure of causal identification. Under the standard assumptions of SUTVA, conditional ignorability, and positivity, causal effects remain identified for any well-defined outcome. Rather, the theorem highlights a measurement problem. Ordinal responses preserve the ordering of latent attitudes but do not reveal the cardinal scale on which those attitudes are measured. Consequently, causal quantities defined on the latent scale require additional measurement assumptions beyond those needed for causal identification itself.

This distinction has immediate implications for empirical practice. Different analyses of ordinal outcomes target different causal quantities and therefore rely on different measurement assumptions. For example, assigning numerical scores to ordinal categories and estimating mean differences implicitly assumes that those scores provide a meaningful cardinal representation of the latent attitude. By contrast, causal quantities defined directly on the observed ordinal responses do not require assumptions about the cardinal spacing between adjacent categories. The next subsection formalizes these probability-based causal estimands and shows how they arise naturally from the observed ordinal response distribution.

2.3. Distributional Causal Estimands

The identification result in Theorem 1 does not imply that causal inference with ordinal outcomes is fundamentally limited. Rather, it suggests that inference should focus on causal quantities defined directly on the observed ordinal responses. Such quantities require no assumptions about the cardinal spacing of the latent attitude and therefore remain invariant to admissible transformations of the latent scale. This perspective is closely aligned with recommendations to report predicted probabilities rather than latent coefficients when interpreting ordinal response models (Hanmer and Ozan Kalkan, 2013). Here I formalize these quantities as causal estimands and show that they are identified under the standard assumptions of causal inference.

The fundamental object is the distribution of potential ordinal outcomes. Let

$$\pi_j(d) = P\left(Y_i(d) = j\right), \qquad j = 0, \ldots, J,$$

denote the probability that a randomly selected individual reports category j under treatment level d.

Under the standard assumptions of SUTVA, conditional ignorability, and positivity, these probabilities are identified by standardization,

$$\pi_j(d) = \mathbb{E}_X\left[P\left(Y_i = j \mid D_i = d, X_i\right)\right],$$

where the expectation is taken over the covariate distribution of the target population. In randomized experiments, this expression simplifies to the observed proportion of respondents falling into category j within each treatment arm.

Proposition 1. *Under SUTVA, conditional ignorability, and positivity, the distribution of potential ordinal outcomes is identified by standardization. Consequently, any causal quantity that is a functional of this* *distribution is also identified.*

This proposition has an important practical implication. Once the conditional ordinal response distribution has been estimated, a wide range of causal quantities become available without fitting additional models. The primary statistical task is therefore estimation of the conditional response probabilities,

$$P(Y_i = j \mid D_i, X_i),$$

which serve as the building blocks for all distributional causal estimands considered in this paper.

The simplest causal quantity is the category-specific treatment effect,

$$\Delta_j = \pi_j(1) - \pi_j(0),$$

which measures how treatment changes the probability that a respondent selects category j. For example, in a survey experiment on support for military intervention, $\Delta_{\text{Strongly Agree}}$ measures the change in the proportion of respondents expressing the strongest level of support.

In many political science applications, however, interest centers on whether respondents exceed a substantively meaningful threshold rather than on a single response category. This motivates cumulative treatment effects,

$$\Delta_{\geq j} = P\left(Y_i(1) \geq j\right) = P\left(Y_i(0) \geq j\right),$$

which quantify changes in the probability of responding at or above category j. For example, a researcher may wish to estimate whether an informational treatment increases the probability that respondents express at least moderate support for a policy. Unlike dichotomizing the outcome before estimation, cumulative treatment effects can be calculated for every threshold after estimating the ordinal response distribution, allowing the full information contained in the ordinal responses to be retained.

These estimands also clarify the relationship between the proposed framework and existing empirical practice. Assigning numerical scores to ordinal categories estimates

$$\mathbb{E} = \left[s(Y_i(1)) - s(Y_i(0))\right],$$

where $s(\cdot)$ denotes the chosen scoring rule. This quantity is well defined for a fixed coding rule but generally does not identify a treatment effect on the underlying latent attitude. Similarly, dichotomizing an ordinal outcome estimates one cumulative treatment effect corresponding to a single threshold while discarding information about the remaining response categories. Estimating the full ordinal response distribution instead makes both category-specific and cumulative treatment effects available simultaneously, allowing researchers to select the quantity most appropriate for their substantive question after estimation rather than before it.

The discussion above highlights an important feature of distributional causal inference. Rather than targeting a single summary measure, the primary object of interest is the conditional distribution of ordinal responses,

$$P(Y_i = j \mid D_i, X_i),$$

from which a wide range of causal quantities can be obtained. Category-specific treatment effects and cumulative treatment effects are the primary estimands considered in this paper because they are directly interpretable and commonly reported in applied political science. More generally, however, any functional of the estimated response distributions may be calculated after estimation, including global measures of distributional change such as the one-dimensional Wasserstein (earth mover's) distance. Consequently, the statistical problem addressed in the remainder of this paper is estimation of the conditional ordinal response distribution rather than any single causal summary.

3. Flexible Estimation of Ordinal Response Distributions

Section 2 established that the causal quantities of interest are functionals of the conditional ordinal response distribution,

$$P(Y_i = j \mid D_i, X_i).$$

The remaining statistical problem is therefore estimation of this distribution. This section first reviews existing estimation approaches and their underlying assumptions before introducing a flexible likelihood-based framework that generalizes classical ordered response models while preserving their probabilistic interpretation.

3.1. Existing Estimation Approaches

The dominant approach to modeling ordinal outcomes in political science is the ordered response model, most commonly ordered logit or ordered probit. These models combine the latent-index representation in Equation (1) with the threshold-crossing measurement model introduced in Section 2.1. The distinction between ordered probit and ordered logit lies only in the assumed distribution of the latent error term, with the former assuming a standard normal distribution and the latter assuming a standard logistic distribution.

Ordered response models remain the workhorses of applied political science because they naturally respect the ordinal structure of survey responses, provide interpretable latent-variable representations, and retain the familiar likelihood-based inferential framework. Moreover, political methodologists have long advocated interpreting these models through predicted probabilities and first differences rather than latent coefficients (Hanmer and Ozan Kalkan, 2013), making them well suited for estimating the distributional causal quantities introduced in Section 2.3.

The validity of these probability estimates, however, depends on the assumptions used to construct the conditional ordinal response distribution. Classical ordered response models impose two principal restrictions. First, the latent error distribution is assumed to belong to a fixed parametric family, typically normal or logistic. Second, the systematic component of the latent outcome is specified through a parametric single-index model. Together, these assumptions determine the shape of the conditional response distribution and therefore the estimated category probabilities.

These assumptions are often reasonable, and when approximately correct ordered response models are statistically efficient. In many survey experiments, however, they may be difficult to justify. Political attitudes are frequently polarized or clustered, producing multimodal latent distributions. Ceiling and floor effects compress responses near the boundaries of Likert scales. Persuasive treatments may affect moderate respondents differently from strong supporters, generating nonlinear treatment-response relationships. Under such conditions, misspecification of either the latent error distribution or the systematic component may distort the estimated category probabilities that form the basis of substantive interpretation.

A substantial literature has relaxed these assumptions. Generalized ordered response models allow threshold-specific effects, while semiparametric estimators relax the distributional assumption on the latent error distribution. Bayesian nonparametric approaches provide additional flexibility through hierarchical modeling. These methods represent important advances, but they typically relax only one component of the model or become computationally demanding in the presence of high-dimensional conditioning variables. The next subsection develops a unified framework that relaxes both sources of misspecification while preserving the likelihood-based structure of conventional ordered response models.

3.2. A Flexible Likelihood-Based Framework

The preceding discussion suggests that the objective is not simply to replace the normal or logistic error distribution with a more flexible alternative. Rather, the goal is to estimate the conditional ordinal response distribution while preserving the probabilistic structure that makes ordered response models attractive. An ideal estimator should satisfy four properties. It should remain likelihood-based, accommodate high-dimensional conditioning variables, recover the full conditional response distribution, and permit varying degrees of structural modeling according to substantive knowledge.

The proposed framework retains the latent outcome representation in Equation (1) but generalizes how its components are modeled. Recall that the latent outcome consists of two parts: the systematic component $h(\cdot)$ and the latent error distribution F. Existing estimators correspond to particular choices of these two components. Ordered logit and ordered probit specify both parametrically, whereas semiparametric estimators typically relax only the latent error distribution.

The proposed framework allows flexibility in either or both components while retaining a common likelihood-based estimation procedure. When substantive theory suggests that treatment and covariates operate through a common latent index, the systematic component may be specified parametrically while the latent error distribution is estimated flexibly. This specification may be viewed as a semiparametric generalization of ordered response models. Alternatively, when the functional relationship between covariates and the latent outcome is itself uncertain, the systematic component may also be estimated nonparametrically. In this way, the same estimation framework spans classical latent-index models, semiparametric ordered response models, and fully flexible conditional distribution estimators without requiring different inferential procedures.

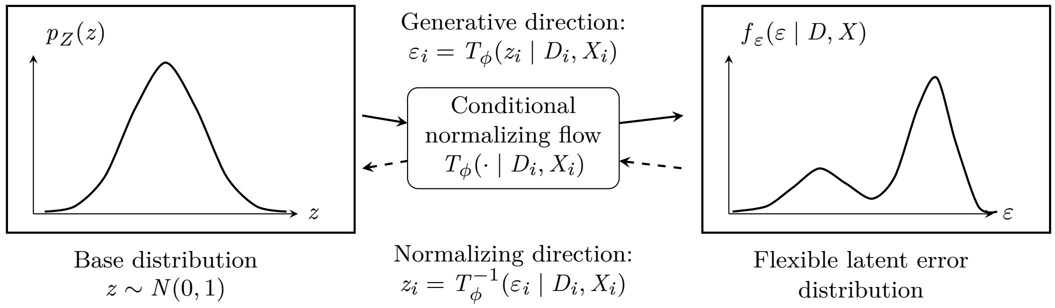
The framework is implemented using conditional normalizing flows to model unknown probability distributions. Unlike kernel-based semiparametric estimators, conditional normalizing flows scale naturally to high-dimensional conditioning variables while retaining exact likelihood evaluation. Consequently, they preserve the familiar inferential machinery of ordered response models, including maximum-likelihood estimation, likelihood-based model comparison, and probabilistic calibration, while substantially relaxing distributional assumptions.

A particularly attractive feature of ordinal response models is that the latent error is scalar. Combined with monotone normalizing flows, this permits exact evaluation of both the latent density and its cumulative distribution function. Consequently, the threshold-crossing likelihood can be evaluated exactly without numerical integration, allowing the proposed estimator to remain computationally tractable while flexibly modeling the latent error distribution.

3.3. Modeling the Latent Error Distribution with Normalizing Flows

The proposed framework replaces the fixed parametric latent error distribution with a flexible conditional density estimator. Rather than assuming that the latent error follows a standard normal or logistic distribution, I estimate its conditional distribution directly from the data while retaining exact likelihood-based inference.

Normalizing flows provide a natural framework for this task. Originally introduced by Rezende and Mohamed (2016), normalizing flows represent an unknown probability distribution through a sequence of in-



Combined with the latent index and thresholds, the estimated error distribution implies conditional ordinal probabilities $P(Y_i = j \mid D_i, X_i)$.

FIGURE 2. Normalizing Flows

A conditional normalizing flow transforms a simple base distribution into a flexible latent error distribution. In the proposed ordered-response framework, this replaces the fixed normal or logistic error assumption while retaining likelihood-based estimation.

vertible transformations of a simple base distribution. Because each transformation is invertible, probability densities remain analytically tractable through the change-of-variables formula, allowing estimation to proceed by maximum likelihood. Since their introduction, normalizing flows have become a standard approach for flexible likelihood-based density estimation (Kobyzev, Prince and Brubaker, 2021; Papamakarios et al., 2021).

Figure 2 illustrates the basic idea. Let

$$z_i \sim p_Z(z)$$

denote a simple base distribution, taken throughout this paper to be the standard normal distribution. A conditional normalizing flow transforms this base distribution into the latent error distribution through an invertible mapping,

$$\varepsilon_i = T_\phi(z_i \mid D_i, X_i),$$

where the transformation is parameterized by ϕ and conditioned on the treatment and covariates. The same transformation may be inverted,

$$z_i = T_\phi^{-1}(\varepsilon_i \mid D_i, X_i),$$

allowing observations to be mapped back to the base distribution. This invertibility is the defining feature of normalizing flows and enables exact likelihood evaluation through the standard change-of-variables formula,

$$f_\varepsilon(\varepsilon_i \mid D_i, X_i) = p_Z(z_i) \left| \det \frac{\partial T_\phi^{-1}(\varepsilon_i \mid D_i, X_i)}{\partial \varepsilon_i} \right|.$$

Unlike kernel density estimators, the transformation is represented by a parametric function that scales naturally to high-dimensional conditioning variables. Throughout this paper I construct $T_\phi(\cdot)$ using monotonic rational-quadratic spline (RQS) transformations (Durkan et al., 2019). RQS flows provide smooth monotone mappings while retaining closed-form evaluation of both the inverse transformation and the Jacobian (Durkan et al., 2019). Because the latent error is one-dimensional, this construction also permits exact evaluation of the implied cumulative distribution function, which is required to compute the threshold probabilities in ordered response models.

The conditional flow is used only to model the latent error distribution. The systematic component of the latent outcome in Equation (1) remains independent of this choice and may therefore be specified at different levels of structural complexity. In the structured specification, the function $h(\cdot)$ is parameterized using a conventional latent-index model, so that flexibility is introduced only through the latent error distribution. This specification may be viewed as a semiparametric generalization of ordered logit and ordered probit. In the fully flexible specification, both the systematic component and the latent error distribution are modeled nonparametrically, allowing the conditional ordinal response distribution to be learned with minimal functional-form assumptions. Both specifications share the same likelihood and estimation procedure, differing only in the amount of structure imposed on the latent outcome model.

3.4. Likelihood-Based Estimation

The latent-index representation in Equation (1), together with the threshold-crossing measurement model introduced in Section 2.1, implies that the probability of observing response category j is

$$\begin{aligned} P(Y_i = j \mid D_i, X_i) &= P\left(\alpha_{j-1} < Y_i^* \le \alpha_j \mid D_i, X_i\right) \\ &= F_\phi\left(\alpha_j - h(X_i) - D_i^\top \tau \mid D_i, X_i\right) - F_\phi\left(\alpha_{j-1} - h(X_i) - D_i^\top \tau \mid D_i, X_i\right), \end{aligned}$$

where $F_\phi(\cdot)$ denotes the conditional cumulative distribution function implied by the normalizing flow introduced in the previous subsection. This expression differs from ordered logit and ordered probit only through the choice of F_ϕ. Rather than assuming a logistic or normal distribution, the latent error distribution is estimated directly from the data.

Also, Unlike general multivariate flow models, the one-dimensional monotone spline construction used here admits exact evaluation of the conditional cumulative distribution function F_ϕ. Consequently, the category probabilities required by the ordered-response likelihood can be computed analytically without numerical integration.

Assuming independent observations, the log-likelihood for the observed ordinal responses is

$$\ell(\Theta) = \sum_{i=1}^{n} \sum_{j=0}^{J} 1(Y_i = j) \log P(Y_i = j \mid D_i, X_i),$$

where

$$\Theta = \tau, h, \phi, \alpha$$

collects all unknown parameters of the latent outcome model, the conditional normalizing flow, and the reporting thresholds.

The parameters are estimated by maximizing this likelihood using stochastic gradient optimization. Because the conditional normalizing flow admits exact evaluation of both the latent density and its cumulative distribution through the invertible transformation, the resulting objective remains a genuine likelihood rather than a surrogate prediction loss. Consequently, estimation retains many of the desirable features of conventional ordered response models, including likelihood-based model comparison and probabilistic calibration, while allowing substantially greater flexibility in the latent error distribution.

The likelihood in Equation (1) encompasses both structured and flexible specifications discussed in the previous subsection. When the systematic component $h(\cdot)$ is parameterized linearly, the estimator may be viewed as a semiparametric generalization of ordered logit and ordered probit. When $h(\cdot)$ is itself modeled nonparametrically, the same likelihood estimates the conditional ordinal response distribution under minimal functional-form assumptions. In either case, the fitted model directly yields the conditional probabilities

$$P(Y_i = j \mid D_i, X_i),$$

which serve as the basis for estimating all causal quantities introduced in Section 2.3. For example, the estimated category-specific treatment effect is obtained by standardization,

$$\widehat{\Delta}_j = \frac{1}{n} \sum_{i=1}^{n} \left[\widehat{P}(Y_i = j \mid D_i = 1, X_i) - \widehat{P}(Y_i = j \mid D_i = 0, X_i) \right],$$

which corresponds to the plug-in estimator of the g-formula (Robins, 1986). Cumulative treatment effects are computed analogously,

$$\widehat{\Delta}_{\geq j} = \frac{1}{n} \sum_{i=1}^{n} \left[\widehat{P}(Y_i \geq j \mid D_i = 1, X_i) - \widehat{P}(Y_i \geq j \mid D_i = 0, X_i) \right].$$

More generally, because the entire conditional response distribution is available after estimation, any functional of that distribution may be computed without fitting additional models. Examples include average predicted probabilities, first differences, stochastic dominance measures, or global summaries of distributional change such as the one-dimensional Wasserstein (earth mover's) distance. The proposed estimator therefore separates estimation from interpretation: the model is fitted once, while substantive quantities of interest are obtained as post-estimation summaries of the estimated response distribution.

The simulation study and empirical applications that follow focus primarily on category-specific and cumulative treatment effects because these quantities are directly interpretable and commonly reported in applied political science. The same fitted model, however, supports a substantially broader class of distributional summaries whenever alternative scientific questions arise. Throughout the simulations and empirical applications, uncertainty is quantified using the nonparametric bootstrap, which naturally propagates estimation uncertainty from the fitted conditional response distribution to all post-estimation functionals.

4. Monte Carlo Simulation

This section evaluates the finite-sample performance of the proposed estimation framework. The objective is not to demonstrate that flexible estimation universally outperforms classical ordered response models. Ordered logit and ordered probit remain statistically efficient when their assumptions are approximately satisfied. Instead, the simulations investigate the conditions under which flexible estimation of the conditional ordinal response distribution improves recovery of the distributional causal effects introduced in Section 2.3. Throughout, causal identification is held fixed so that differences in performance arise solely from the estimation procedure.

4.1. Simulation Design

Each simulation begins by generating a synthetic population of one million observations from the latent-index model in Equation (1). The latent outcome is converted into an observed ordinal response through the threshold-crossing measurement model introduced in Section 2.1, with common reporting thresholds fixed across treatment conditions. The resulting population serves as the ground truth throughout the Monte Carlo experiment. Because the population is sufficiently large, the true category-specific and cumulative treatment effects can be computed with negligible Monte Carlo error before sampling begins. Consequently, all reported estimation error reflects finite-sample performance rather than uncertainty in the reference values.

Treatment assignment is completely randomized, ensuring that the identification assumptions developed in Section 2 hold by construction. For each data-generating process, I draw 100 independent random samples of sizes $n \in 500, 1000, 2000$ from the population. Each sampled dataset is analyzed independently, and performance measures are aggregated across Monte Carlo replications.

The simulation considers six data-generating processes designed to isolate different departures from the assumptions underlying classical ordered response models. Two benchmark settings generate latent errors from the normal and logistic distributions, under which ordered probit and ordered logit are correctly specified, respectively. The remaining settings introduce skewed latent errors, polarized (multimodal) latent attitudes, heteroskedastic latent errors, and nonlinear latent response functions. Together these scenarios represent empirical features frequently encountered in survey experiments while allowing the consequences of different modeling assumptions to be examined separately.

Five estimators are compared. The first is an empirical distribution estimator that estimates category probabilities directly from observed treatment-arm frequencies without imposing a statistical model. The second and third are ordered probit and ordered logit. The remaining estimators implement the proposed

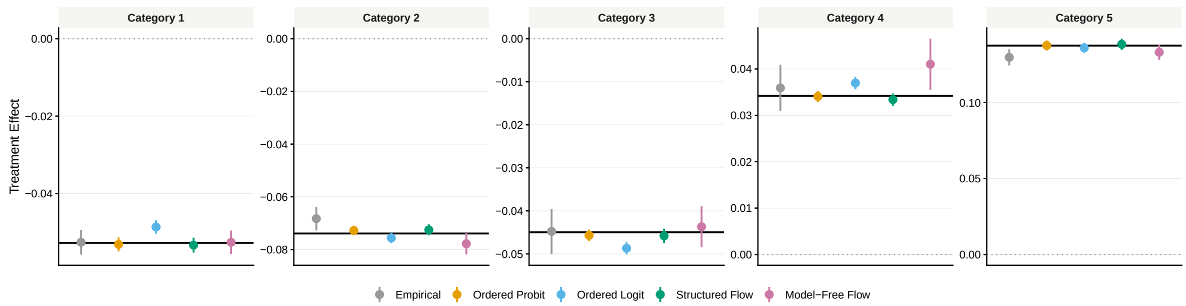


FIGURE 3. Single Index Model with Normal Error

Note: This figure reports the estimated category-specific treatment effects with the corresponding population quantities. Points denote Monte Carlo averages across 100 replications, error bars indicate 95% Monte Carlo confidence intervals, and the horizontal line marks the true population treatment effect.

framework in two forms. The structured flow specification retains the linear latent-index model while estimating the latent error distribution flexibly. The model-free specification estimates both the systematic component and the latent error distribution nonparametrically. To facilitate stable optimization, both flow estimators are initialized using the ordered probit estimates before likelihood maximization.

Performance is evaluated using bias, absolute bias, and root mean squared error (RMSE) for the category-specific and cumulative treatment effects introduced in Section 2.3. These quantities are obtained by regression standardization from each fitted model. Because all estimators target the same causal estimands under identical identification assumptions, differences in performance reflect only the statistical consequences of the modeling assumptions imposed by each estimation procedure.

4.2. Simulation Results

The benchmark simulation in Figure 3 provides an important validation of the proposed framework. Under normally distributed latent errors and a linear latent-index model, where ordered probit is correctly specified, all model-based estimators recover the category-specific treatment effects with little bias . The structured flow estimator performs comparably to ordered probit despite relaxing the parametric assumption on the latent error distribution, indicating that the additional flexibility incurs little efficiency loss when the conventional model is correctly specified.

The advantages of flexible estimation become apparent once the latent response mechanism departs from the assumptions of classical ordered response models. Figure 4 demonstrates that under the polarized mixture distribution, ordered logit and ordered probit systematically underestimate treatment effects in the extreme response categories, reflecting their inability to accommodate multimodal latent attitudes. The structured

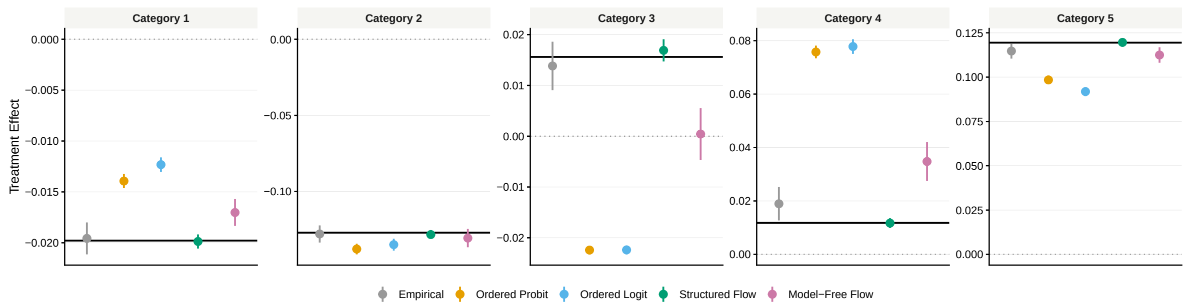


FIGURE 4. Single Index Model with Bimodal Error

Note: This figure reports the estimated category-specific treatment effects with the corresponding population quantities. Points denote Monte Carlo averages across 100 replications, error bars indicate 95% Monte Carlo confidence intervals, and the horizontal line marks the true population treatment effect.

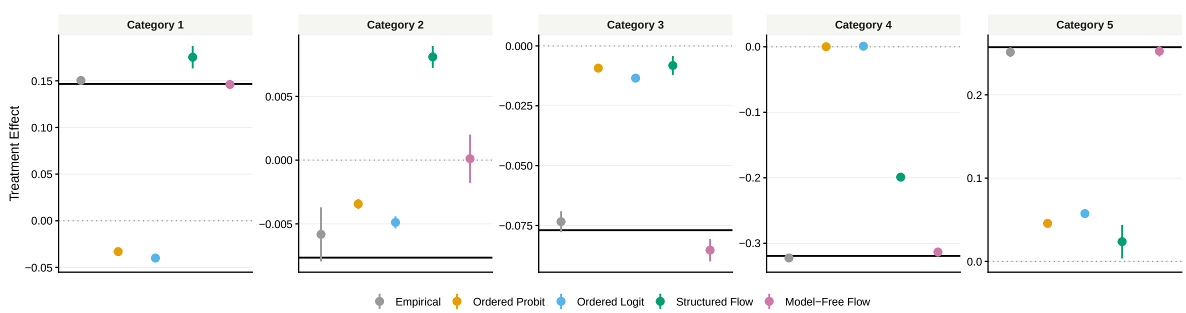


FIGURE 5. Single Index Model with heteroskedastic Error

Note: This figure reports the estimated category-specific treatment effects with the corresponding population quantities. Points denote Monte Carlo averages across 100 replications, error bars indicate 95% Monte Carlo confidence intervals, and the horizontal line marks the true population treatment effect.

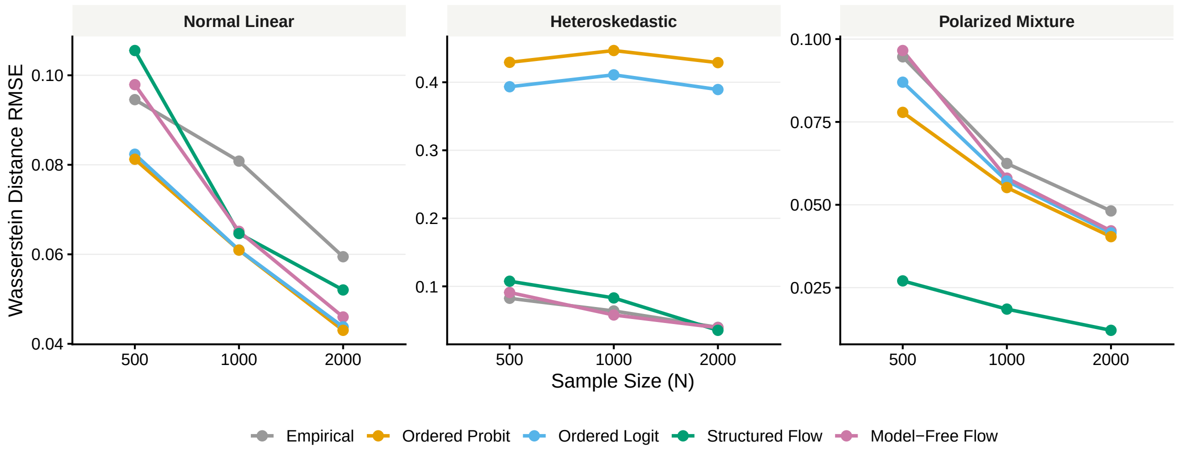


FIGURE 6. Asymptotic Convergence (RMSE) in Wasserstein Distance

Note: This figure reports recovery of the entire conditional response distribution through the RMSE of the Wasserstein distance between the estimated and true distributions.

flow estimator closely tracks the population treatment effects across nearly all response categories, while the model-free specification reduces some of the bias but exhibits greater variability.

The heteroskedastic design in Figure 5 presents a more demanding form of misspecification because treatment changes both the location and dispersion of the latent outcome. Here the parametric ordered response models exhibit substantial bias across multiple response categories, particularly in the upper tail of the response distribution. In contrast, the structured flow estimator continues to recover the treatment effects with comparatively small bias, demonstrating that flexibly modeling the latent error distribution substantially improves estimation when the proportional-scale assumption is violated.

Figure 6 summarizes these findings by examining recovery of the entire conditional response distribution rather than individual treatment effects. The Wasserstein distance measures the discrepancy between the estimated and true response distributions, with smaller values indicating more accurate recovery. Across all sample sizes, estimation error decreases as expected, but the relative ordering of the estimators remains stable. The structured flow estimator consistently achieves the smallest Wasserstein error under heteroskedastic and polarized latent response mechanisms while remaining competitive under the benchmark setting. These results indicate that the proposed framework improves estimation not only for individual category-specific effects but also for the overall ordinal response distribution.

Taken together, the simulations suggest a clear practical conclusion. When the assumptions underlying ordered response models are approximately correct, the proposed estimator performs comparably to conventional parametric methods. When these assumptions are violated, however, flexibly estimating the latent error distribution substantially improves recovery of both the conditional ordinal response distribution and the resulting distributional causal effects. Perhaps more importantly, the simulations indicate that retaining the latent-index structure while relaxing the latent error distribution provides a better balance between robustness and statistical efficiency than fully nonparametric estimation.

5. Empirical Applications

The simulation study examined the finite-sample performance of the proposed estimator under controlled data-generating processes where the true response distribution is known. This section investigates its practical implications using two published survey experiments. The applications were selected because both analyze ordinal outcomes using standard causal designs, yet they represent contrasting empirical settings. In one application, the estimated treatment effects are largely stable across modeling assumptions, whereas in the other, relaxing the parametric assumptions underlying conventional ordered response models leads to substantively different conclusions. Throughout, I report category-specific average marginal effects obtained by regression standardization together with nonparametric bootstrap confidence intervals.

5.1. Tomz and Weeks (2020): Human Rights and Support for Military Intervention

I first revisit the survey experiment of Tomz and Weeks (2020), which examines how information about foreign governments' human rights records influences public support for military intervention. Respondents were randomly assigned to treatment conditions describing the target country's human rights practices and subsequently reported their level of support for military action using a five-category ordinal response scale. The original analysis estimated treatment effects using conventional approaches based on ordinal response models.

Figure 7 compares the estimated category-specific treatment effects obtained from the empirical distribution estimator, ordered probit, ordered logit, and the two proposed flow-based estimators. Overall, the estimated treatment effects are remarkably consistent across estimation methods. All five approaches indicate that the treatment shifts respondents away from the highest levels of support for military intervention and toward lower response categories, with only modest differences in the estimated magnitudes. The structured flow estimator closely tracks the estimates produced by the classical ordered response models, while the model-free specification yields slightly greater uncertainty but no substantial change in the qualitative pattern of results.

This application illustrates an important point. Flexible estimation is not intended to replace conventional ordered response models regardless of context. Rather, it provides a robustness check against restrictive

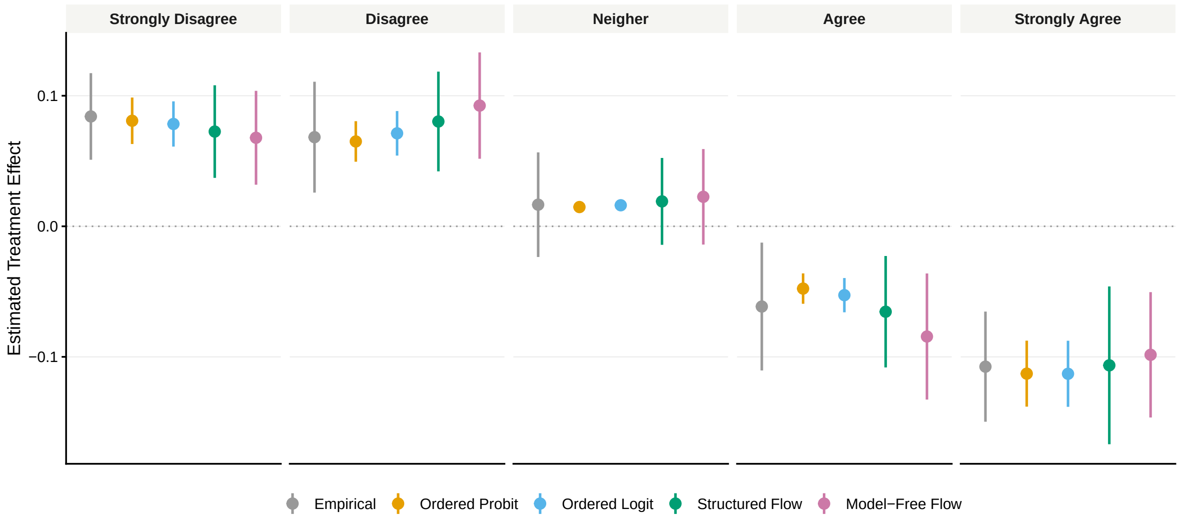


FIGURE 7. Category Specific Effects on Tomz and Weeks (2020)

Note: This figure reports the estimated category-specific treatment effects in replication of Tomz and Weeks (2020). Points denote category specific effect estimates and error bars indicate Bootstrapped 95% Confidence Intervals.

modeling assumptions. In this case, the substantive conclusions appear largely insensitive to the assumed latent error distribution, suggesting that the conventional ordered response models provide an adequate approximation for the observed data. The proposed estimator therefore reproduces the existing substantive findings while demonstrating that they are robust to substantially weaker distributional assumptions.

5.2. Mattingly et al. (2025): Authoritarian Propaganda and Public Opinion

I next revisit the survey experiment of Mattingly et al. (2025), which investigates the persuasive effects of authoritarian propaganda on public opinion. Respondents were randomly assigned to different informational treatments and subsequently evaluated the target using a six-category ordinal response scale. As in the previous application, the original analysis relied on conventional ordinal response methods to summarize treatment effects.

Figure 8 presents the estimated category-specific treatment effects across the five estimators. Unlike the Tomz application, the estimated effects now exhibit substantial disagreement across modeling assumptions. The empirical estimator, ordered probit, ordered logit, and the two flow-based estimators produce noticeably different estimates for several response categories, particularly in the middle and upper portions of the ordinal scale. While all methods identify a positive treatment effect, they imply different patterns of movement across response categories and therefore different substantive interpretations of how the treatment changes respondents' opinions.

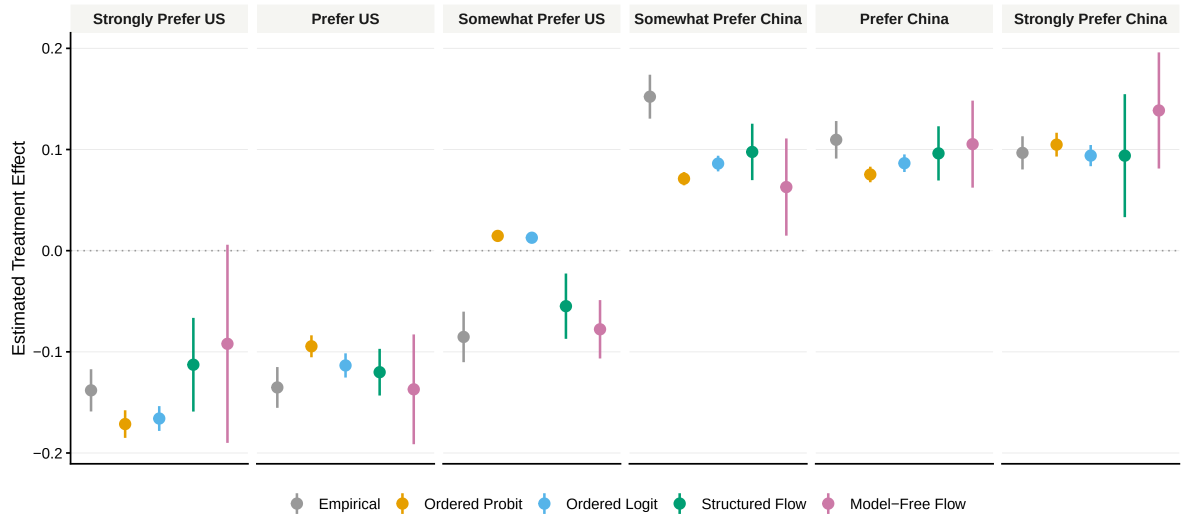


FIGURE 8. Category Specific Effects on Mattingly et al. (2025)

Note: This figure reports the estimated category-specific treatment effects in replication of Mattingly et al. (2025). Points denote category specific effect estimates and error bars indicate Bootstrapped 95% Confidence Intervals.

The disagreement among estimators is not uniform across the response distribution. Rather than differing in the overall direction of the treatment effect, the models primarily disagree about how probability mass is redistributed across adjacent response categories. Relative to ordered probit and ordered logit, the structured flow estimator attributes more of the treatment effect to movement among the intermediate response categories while assigning less probability to shifts concentrated in the upper tail of the distribution. By contrast, the estimated effects for the extreme response categories are comparatively similar across methods. The fully model-free specification exhibits a broadly similar pattern but with wider confidence intervals, reflecting the additional uncertainty associated with estimating the systematic component nonparametrically. These results closely mirror the simulation findings, suggesting that retaining the latent-index structure while flexibly modeling the latent error distribution provides a better balance between robustness and statistical efficiency than either fully parametric or fully nonparametric estimation.

This application illustrates the practical value of flexible distributional modeling. The differences among estimators do not arise because the causal effect itself is unidentified. Rather, they arise because different assumptions about the latent response distribution imply different estimates of the conditional ordinal response probabilities. When the parametric assumptions underlying ordered response models are difficult to justify, these differences may be large enough to alter substantive conclusions about where in the response distribution the treatment has its greatest impact.

6. Conclusion

Ordinal outcomes are among the most common outcome measures in political science, yet they present distinctive challenges for causal inference because the observed response categories reveal only the ordering of an underlying latent attitude. This paper revisited the classical identification result that latent treatment effects are identified only up to location and scale and argued that probability-based causal quantities defined on the observed ordinal responses provide a natural target for empirical analysis. Building on this perspective, I developed a flexible likelihood-based framework that estimates the conditional ordinal response distribution while relaxing the restrictive distributional assumptions imposed by conventional ordered response models.

The proposed framework retains the familiar latent-variable representation underlying ordered logit and ordered probit while replacing the fixed latent error distribution with a conditional normalizing flow. This construction preserves exact likelihood-based inference and accommodates both structured latent-index models and fully flexible conditional response distributions within a common estimation procedure. Rather than treating flexibility as an end in itself, the framework allows researchers to relax only those assumptions that are most difficult to justify while preserving the interpretability and statistical efficiency of classical ordinal response models.

The simulation study and empirical applications illustrate the practical consequences of this approach. When the assumptions underlying ordered response models are approximately correct, the proposed estimator performs comparably to conventional methods. When the latent response distribution departs from these assumptions, however, flexible estimation substantially improves recovery of both the conditional ordinal response distribution and the resulting distributional causal effects. The empirical applications further demonstrate that the sensitivity of substantive conclusions to distributional assumptions varies across studies. In some settings, conventional ordered response models provide essentially the same conclusions as the proposed estimator. In others, relaxing the latent error distribution meaningfully changes where in the ordinal response distribution treatment effects are estimated to occur.

More broadly, this paper argues that the conditional ordinal response distribution, rather than any single latent coefficient, should be viewed as the primary statistical object for causal inference with ordinal outcomes. Once this distribution has been estimated, a wide range of scientifically meaningful causal summaries, including category-specific effects, cumulative treatment effects, and other distributional functionals, follow naturally through standardization. This perspective separates the statistical problem of estimating the response distribution from the substantive problem of selecting the causal quantity most appropriate for a particular research question.

Several extensions remain for future work. The framework developed here assumes a common reporting function across respondents, whereas heterogeneous reporting behavior and measurement error remain important challenges for many survey instruments. Extending the proposed approach to multiple-item ordinal outcomes, integrating it with doubly robust estimation for observational studies, and developing formal specification diagnostics for parametric ordinal response models represent promising directions for future research.

As ordinal outcomes continue to play a central role in political science, the goal need not be to replace familiar ordered response models, but to extend them. Flexible likelihood-based estimation offers a practical way to preserve the strengths of classical ordinal models while reducing their dependence on restrictive distributional assumptions, thereby providing more robust causal inference when the underlying response process is uncertain.

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